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BSCS-1E

Digital Logic Design - Theory [Assignment #1]

Q1. Convert $[1073]_{10}$ into a Binary number.

2	1073
2	536 - 1
2	268 - 0
2	134 - 0
2	67 - 0
2	33 - 1
2	16 - 1
2	8 - 0
2	4 - 0
2	2 - 0
	1 - 0

$\therefore [010000110001]_2$

Q2. Convert $[81]_{10}$ to Binary.

	128	64	32	16	8	4	2	1
=	0	1	0	1	0	0	0	1

$$\begin{array}{r} 64 \\ 16 \\ 1 \\ \hline 81 \end{array}$$

$\therefore [01010001]_2$

Q3. Convert $[27.315]_{10}$ to a binary

128	64	32	16	8	4	2	1	.	0.5	0.25	0.125	0.0625	0.03125	0.015625
0	0	0	1	1	0	1	1	.	0	1	0	1	0	0

$\therefore [11011.0101]_2$

Note:- this is 27.3125 equivalent answer because to get 27.315 we will need to go to a much small binary values, which is not possible to show on a paper. Therefore this is the closest answer i could get to

Q4a)
$$\begin{array}{r} 11010 \\ +\ 11100 \\ \hline ?? \end{array}
\qquad
\begin{array}{rl} 00011010 & \rightarrow 26 \\ +\ 00011100 & \rightarrow 28 \\ \hline 00110110 & \rightarrow 54 \end{array}$$

$\therefore [00110110]_2$

b) $(101011)_2 + (110101)_2 = ?$

$$\begin{array}{rl} 00101011 & \rightarrow 43 \\ +\ 00110101 & \rightarrow 53 \\ \hline 01100000 & \rightarrow 96 \end{array}$$

$\therefore [01100000]_2$

Q5a) $(101110)_2 - (100100) = ?$

$$\begin{array}{rl} 101110 & \rightarrow 46 \\ -\ 100100 & \rightarrow 36 \\ \hline 001010 & \rightarrow 10 \end{array}$$

$\therefore [001010]_2$

b) $(1001100)_2 - (110)_2 = ?$

$$\begin{array}{rl} 01001100 & \rightarrow 76 \\ -\ 00000110 & \rightarrow 6 \\ \hline 01000110 & \rightarrow \cancel{7}0 \end{array}$$

$\therefore [01000110]_2$

Q6) Convert hexa decimal to octal

a) $(FA25)_{16}$

Hexa-decimal to Binary

8 4 2 1	8 4 2 1	8 4 2 1	8 4 2 1
1 1 1 1	1 0 1 0	0 0 1 0	0 1 0 1

= 1111101000100101

Binary to Octal

	4 2 1	4 2 1	4 2 1	4 2 1	4 2 1	4 2 1
	0 0 1	1 1 1	1 0 1	0 0 0	1 0 0	1 0 1
=	1	7	5	0	4	5

$\therefore [175045]_8$

Q6b) $(F920)_{16}$

Hexa-decimal to Binary

8421	8421	8421	8421
1111	1001	0010	0000

= 1111100100100000

Binary to Octal

421	421	421	421	421	421
001	111	100	100	100	000
= 1	7	4	4	4	0

$\therefore [174440]_8$

Q6c) $(1100)_{16}$

8421	8421	8421	8421
0001	0001	0000	0000

= 0001000100000000

Binary to Octal

421	421	421	421	421	421
000	001	000	100	000	000
= 0	1	0	4	0	0

$\therefore [010400]_8$

Q7) Convert Octal to Hexa-decimal

a) $(777)_8$

Octal to Binary

421	421	421
111	111	111

= 111111111

Binary to Hexa-decimal

8421	8421	8421
0001	1111	1111

$\Rightarrow$ 1FF

$\therefore [1FF]_{16}$

Q7b) $(123)_8$

Octal to Binary :-

4 2 1	4 2 1	4 2 1
0 0 1	0 1 0	0 1 1

= 001010011

Binary to Hexa-decimal

8 4 2 1	8 4 2 1	8 4 2 1
0 0 0 0	0 1 0 1	0 0 1 1

= 0 5 3

$\therefore [53]_{16}$

Q7c) $(635)_8$

Octal to Binary :-

4 2 1	4 2 1	4 2 1
1 1 0	0 1 1	1 0 1

= 110011101

Binary to Hexa-decimal

8 4 2 1	8 4 2 1	8 4 2 1
0 0 0 1	1 0 0 1	1 1 0 1
= 1	9	D

$\therefore [19D]_{16}$

Q8) Express each decimal number as an 8-bit sign magnitude binary number.

a) −83

	128	64	32	16	8	4	2	1	
=	0	1	0	1	0	0	1	1	} + 83 binary form

~~Now add its 2's complement~~

Now add sign bit and magnitude, to make it sign magnitude binary number.

Sign bit :- Since number is -ve, the sign bit is 1.

Magnitude :- $(83)_{10}$ = 1010011 } 7-bit binary

combine sign bit & magnitude :- $\underbrace{1}_{\text{MSB}}\ \underbrace{1010011}_{\text{magnitude}}$

$\therefore [11010011]_2$

Q8b) $(+101)_{10}$

128	64	32	16	8	4	2	1	
0	1	1	0	0	1	0	1	} +101 binary form

sign bit :- Since number is positive the sign bit is 0

magnitude :- $(101)_{10}$ = 1100101 } 7-bit binary

combine sign bit & magnitude :- $\underbrace{0}_{\text{MSB}}\underbrace{1100101}_{\text{magnitude}}$

$\therefore [01100101]_2$

Q8c) $(-114)_{10}$

128	64	32	16	8	4	2	1	
0	1	1	1	0	0	1	0	} 114 binary form

sign bit :- Since number is negative the sign bit is 1

magnitude :- $(114)_{10}$ = 1110010 } 7-bit binary

combine sign bit & magnitude :- $\underbrace{1}_{\text{MSB}}\underbrace{1110010}_{\text{magnitude}}$

$\therefore [11110010]_2$

Q4a) Express each decimal number in ~~binary~~ as ~~a~~n 8-bit number in the 1's compliment form.

a) $(-66)_{10}$

~~[illegible]~~

magnitude:-

128	64	32	16	8	4	2	1
0	1	0	0	0	0	1	0

= 01000010 } 8-bit binary

00111101 → 1's compliment

Apply 1's compliment ⇒ 01000010

$[10111101]_2$ ans

Q9b) $(+116)_{10}$

128	64	32	16	8	4	2	1
0	1	1	1	0	1	0	0

for +ve numbers, the 1's complement is the same as the original binary.

$\therefore [01110100]_2$

Q9c) $(-99)_{10}$

convert +99 to 8-bit binary.

128	64	32	16	8	4	2	1	
0	1	1	0	0	0	1	1	
1	0	0	1	1	1	0	0	} 1's compliment

$\therefore [10011100]_2$

Q10) Express each decimal number as an 8-bit number in the 2's compliment form.

a) $(-59)_{10}$

128	64	32	16	8	4	2	1	
0	0	1	1	1	0	1	1	} +59
1	1	0	0	0	1	0	0	} 1's compliment
+ 0	0	0	0	0	0	0	1	} 2's compliment (add 1)
1	1	0	0	0	1	0	1	

$\therefore [11000101]_2$

Q10b) $(+102)_{10}$

128	64	32	16	8	4	2	1
0	1	1	0	0	1	1	0

for +ve numbers the 1's compliment and 2nd compliment remains safe.

$\therefore [01100110]_2$

Q10c) $(-126)_{10}$

	128	64	32	16	8	4	2	1	
	0	1	1	1	1	1	1	0	} 126 binary format
	1	0	0	0	0	0	0	1	} 1's compliment
+	0	0	0	0	0	0	0	1	
	1	0	0	0	0	0	1	0	

$\therefore [10000010]_2$

Q11) Determine the decimal value of each signed binary number in the sign-magnitude form:

a) $(10011101)_2$

$\underbrace{1}_{\text{sign (1 represent negative number)}}\underbrace{0011101}_{\text{magnitude [7-bit binary]}}$

convert magnitude

64	32	16	8	4	2	1
0	0	1	1	1	0	1

$= 16 + 8 + 4 + 1$

$= 29$ } add -ve sign

$= -29$

$\therefore [-29]_{10}$ is equivalent to $(10011101)_2$ sign - magnitude form

b) $(01110100)_2$

$\underbrace{0}\ \underbrace{1110100}$

↳ magnitude (7-bit binary)

↳ signed bit (0 represent positive number)

convert magnitude

64	32	16	8	4	2	1
1	1	1	0	1	0	0

$= 64 + 32 + 16 + 4$

$= 116$ } add in the +ve sign

$= +116$

$\therefore (+116)_{10}$ is equivalent to $(01110100)_2$ sign-magnitude form

c) $(10111011)_2$

$\underbrace{1}\ \underbrace{0111011}$

↳ magnitude (7-bit binary)

↳ signed bit (1 represent negative number)

convert magnitude

64	32	16	8	4	2	1
0	1	1	1	0	1	1

$= 32 + 16 + 8 + 2 + 1$

$= 59$ } add -ve sign

$= -59$

$\therefore [-59]_{10}$ is equivalent to $(10111011)_2$ sign magnitude form

Q12) Determine the decimal value of each signed binary number in the 1's compliment form.

a) $(10111001)_2$

The most-significant-bit is 1 so it is a negative number

Doing reverse of 1's compliment to get original value

10111001

01000110 } 1's compliment (original)

Convert Binary to Decimal

128 64 32 16 8 4 2 1
0 1 0 0 0 1 1 0

= 64+4+2

= -70 } add negative sign due to M.S.B being 1

∴ $[-70]_{10}$

Q12b) $(01100100)_2$

As the sign bit is 0 it means the number is positive and that no reverse inversion to be done. Simply conversion from binary to decimal.

Conversion :- 128 64 32 16 8 4 2 1
0 1 1 0 0 1 0 0

= 64 + 32 + 4

= +100

∴ $[+100]_{10}$

Q12c) $(10111101)_2$

The most-significant bit is 1 so it's a negative number.

Applying 1's compliment

10111101
01000010 } 1's compliment

Convert Binary to decimal :-

128 64 32 16 8 4 2 1
0 1 0 0 0 0 1 0

= -66 } add negative sign

∴ $[-66]_{10}$

Q13) A system uses 8-bit two's complement representation for signed numbers. What is the decimal equivalent of the following binary numbers?

a) $(10101100)_2$

The sign-bit is 1 so it's negative decimal number.

Conversion from Binary to Decimal :

-128	64	32	16	8	4	2	1
= 1	0	1	0	1	1	0	0

$= -128 + 32 + 8 + 4$

$= \cancel{-86}\ -84$

$\therefore (-84)_{10}$

b) $(01111001)_2$

The sign-bit is 0 so it's positive decimal number.

Conversion from binary to decimal :-

-128	64	32	16	8	4	2	1
= 0	1	1	1	1	0	0	1

$= 64 + 32 + 16 + 8 + 1$

$= +121$

$\therefore (+121)_{10}$

c) $(11110000)_2$

The sign-bit is 1 so it's negative decimal number.

Conversion from binary to decimal :-

-128	64	32	16	8	4	2	1
= 1	1	1	1	0	0	0	0

$= -128 + 64 + 32 + 16$

$= -16$

$\therefore (-16)_{10}$

Q14a) Convert each pair of decimal numbers to binary and add using the 2's compliment form (8 bit representation).

a) −38 and −27

Converting −38 to Binary

128	64	32	16	8	4	2	1	
0	0	1	0	0	1	1	0	} +38
1	1	0	1	1	0	0	1	} 1's compliment
+ 0	0	0	0	0	0	0	1	} 2's compliment
1	1	0	1	1	0	1	0	→ −38

Converting −27 to Binary

128	64	32	16	8	4	2	1	
0	0	0	1	1	0	1	1	} +27
1	1	1	0	0	1	0	0	} 1's compliment
+ 0	0	0	0	0	0	0	1	} 2's compliment
1	1	1	0	0	1	0	1	→ −27

add

```
 1
  11011010  → −38
  11100101  → −27
  --------    ---
  10111111  → −65
```

∴ $(10111111)_2$

14b) 59 and −39

Converting −39 to Binary

128	64	32	16	8	4	2	1	
0	0	1	0	0	1	1	1	} +39
1	1	0	1	1	0	0	0	} 1's compliment
+ 0	0	0	0	0	0	0	1	} 2's compliment
1	1	0	1	1	0	0	1	

Converting 59 to Binary

128	64	32	16	8	4	2	1
0	0	1	1	1	0	1	1

add

$$\begin{array}{rl} 00111011 & \rightarrow +59 \\ +\ 11011001 & \rightarrow -39 \\ \hline 00010100 & +20 \end{array}$$

$\therefore (00010100)_2$

14c) −58 and 65

128	64	32	16	8	4	2	1	
0	0	1	1	1	0	1	0	→ +58
1	1	0	0	0	1	0	1	} 1's compliment
+ 0	0	0	0	0	0	0	1	} 2's compliment
1	1	0	0	0	1	1	0	

Converting 65 to Binary

128	64	32	16	8	4	2	1
0	1	0	0	0	0	0	1

add

$$\begin{array}{rl} 01000001 & \rightarrow +65 \\ +\ 11000110 & \rightarrow -58 \\ \hline 00000111 & +07 \end{array}$$

$\therefore (00000111)_2$

14d) −102 and −85

Converting −102 to Binary

128	64	32	16	8	4	2	1	
0	1	1	0	0	1	1	0	} +102
1	0	0	1	1	0	0	1	} 1's compliment
+ 0	0	0	0	0	0	0	1	} 2's compliment
1	0	0	1	1	0	1	0	

Converting -85 to Binary

128	64	32	16	8	4	2	1	
0	1	0	1	0	1	0	1	} +85
1	0	1	0	1	0	1	0	} 1's compliment
+ 0	0	0	0	0	0	0	1	} 2's compliment
1	0	1	0	1	0	1	1	

add

```
  10011010   → -102
+ 10101011   → - 85
-----------    -----
 100100101     -187
```

1 ↳ overflow ignore

∴ $(00100101)_2$ the output is +64 due to overflow rather than -187 as it was suppose to be.

Q15) What is Binary Coded Decimal (BCD), and how does it differ from regular binary representation?

Ans) BCD is a binary representation of decimal numbers where each digit of the decimal number is stored /encoded seperately into a 4-bit binary number. Whereas in Regular Binary the entire ~~de~~ number is converted to binary as a whole.

- The BCD has a range that each 4-bit group represents a decimal digit (0-9), so it can't represent values greater than 9 in a single group. Wheras in Regular Binary the entire number is represented hence wider range of values are stored.
- BCD is less efficient in terms of storage and computation compared to regular binary. Whereas Regular Binary is more efficient for arthimetic operations and storage.

Eg:- Decimal : 39

BCD :- 0011 1001

Regular Binary :- 100111

Q16) Analyse your surroundings and think about applications of BCD, where it is being used?

Ans) BCD is used in digital displays like calculators, digital clocks, LED/LCD displays, in digital measurement instrument like weighing scales, temperature sensors and etc.
It is used in such location as BCD simplifies the conversion of binary data to human-readable decimal digits for display and it also provides accurate decimal readings for measurements

Q17) Convert the following decimal numbers to BCD

a) $(57)_{10}$

converting 5	converting 7
8 4 2 1	8 4 2 1
0 1 0 1	0 1 1 1

$\therefore (0101\ 0111)_{BCD}$

b) $(109)_{10}$

converting 1	converting 0	converting 9
8 4 2 1	8 4 2 1	8 4 2 1
0 0 0 1	0 0 0 0	1 0 0 1

$\therefore (0001\ 0000\ 1001)_{BCD}$

Q18) Add the following numbers after conversion to BCD

a) 7 + 9

converting 7	converting 9
8 4 2 1	8 4 2 1
0 1 1 1	1 0 0 1

add them

1 1 1

1001 → 9

0111 → 7

1 0000 } exceeds 9

0110 } add 6

a'

add them

$$\begin{array}{r} 1001 \rightarrow 9 \\ +\ 0111 \rightarrow 7 \\ \hline 10000 \end{array}$$ } invalid as it exceeds 4 bit ∴ need to add 6

$$\begin{array}{r} 0001\ 0000 \\ +\ \quad 0110 \\ \hline 0001\ 0110 \end{array}$$ } adding 6

∴ $(0001\ 0110)_{BCD}$ which is equivalent to $(16)_{10}$

Q18b) 25 + 58

converting 2

8	4	2	1
0	0	1	0

converting 5

8	4	2	1
0	1	0	1

⇒ (0010 0101)

converting 5

8	4	2	1
0	1	0	1

converting 8

8	4	2	1
1	0	0	0

⇒ (0101 1000)

add them

$$\begin{array}{r} 0101\ 1000 \rightarrow 58 \\ +\ 0010\ 0101 \rightarrow 25 \\ \hline 0111\ 1101 \\ +\ 0000\ 0110 \\ \hline 1000\ 0011 \end{array}$$ } adding 6

∴ $(1000\ 0011)_{BCD}$ which is equivalent to $(83)_{10}$

c) 76 + 84

converting 7

8	4	2	1
0	1	1	1

converting 6

8	4	2	1
0	1	1	0

⇒ (0111 0110)

converting 8

8	4	2	1
1	0	0	0

converting 4

8	4	2	1
0	1	0	0

⇒ (1000 0100)

add them

1000 0100 → 84
+ 0111 0110 → 76

1 1111 1010
+ 0110 + 0110 } adding 6

0001 0110 0000

∴ $(0001\ 0110\ 0000)_{BCD}$ is equivalent to $(160)_{10}$

18 d) 89 + 68

converting 8
8 4 2 1
1 0 0 0

converting 9
8 4 2 1
1 0 0 1

⇒ (1000 1001)

converting 6
8 4 2 1
0 1 1 0

converting 8
8 4 2 1
1 0 0 0

⇒ (0110 1000)

add them

1000 1001 → 89
+ 0110 + 1000 → 68

0001 1111 10001 } invalid both terms > 9
+ 0110 + 0110

0001 0101 1 0111

Note:- The circled one is sent as a carry

∴ $(0001\ 0101\ 0111)_{BCD}$ is equivalent to $(157)_{10}$

Q19) Determine which of the following even parity codes are in error.

a) 110011001

The no. of 1 in this is 5 which is odd, hence the code is in error.

b) 101111101000101 0

The no. of 1 in this is 10 which is even, hence the code is correct

Q19c) 010101110

The no. of 1's in this is 5 which is odd, hence the code is in error.

Q19d) 0111000100101101

The no. of 1 in this is 8 which is even, hence the code is correct.

Q20) Assign the proper odd parity bits to the following code groups.

a) 0110

The no. of 1 is even, hence add 1 bit to it

∴ 10110

b) 101101

The no. of 1 is even, hence add 1 parity bit to it.

∴ 1101101

c) 101101011111

The no. of 1 is odd, hence add parity bit of 0 to it.

∴ 0101101011111

d) 100011100101

The no. of 1 is even, hence add parity bit of 1 to it.

∴ 1100011100101.